Lecture 27 - Course Review

$$A = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 2 & 4 & 1 & 4 \\ 3 & 6 & 3 & 9 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 3 & 6 \end{bmatrix} \rightarrow \underbrace{\begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}}_{U}$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 3 & 1 \end{bmatrix}$$

$\underline{\text{rank}(A) = 2}$

Is $Ax=b$ consistent for all $b \in \mathbb{R}^3$? $\underline{\underline{\text{NO}}}$

null(A): $x_1 = -2x_2 - x_4$

$x_3 = -2x_4$

$$x = \begin{bmatrix} -2x_2 - x_4 \\ x_2 \\ -2x_4 \\ x_4 \end{bmatrix} = x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -1 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$

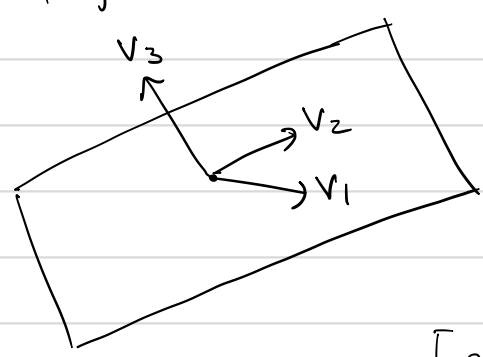
Basis for null(A)

b = sum of all columns of A

Complete solution of $Ax=b$?

$$x = \underbrace{\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}}_{x_p} + \underbrace{x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -1 \\ 0 \\ -2 \\ 1 \end{bmatrix}}_{x_{null}}$$

P: 3x3 matrix, projects onto $x_1 + 2x_2 + 3x_3 = 0$



$\lambda_1 = \lambda_2 = 1$

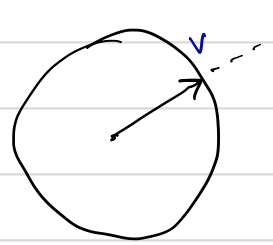
$\lambda_3 = 0$

$x_1 = -2x_2 - 3x_3$

$$x = x_2 \underbrace{\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}}_{v_1} + x_3 \underbrace{\begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}}_{v_2}$$

$$v_3 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \qquad v_3^T v_1 = 0 = v_3^T v_2.$$

Is



a subspace of $\mathbb{R}^2$? $\underline{\text{NO}}$.

$x_1^2 + x_2^2 = 1$